

from the strategic synergy between Lava's R&D, Saankhya Labs' (a Tejas Networks company) indigenous semiconductor innovation, and MediaTek's reliable mobile platforms. Saankhya, a pioneer in software-defined radio (SDR) in India, played a key role with its SL3000 chipset, designed to receive terrestrial digital broadcasts—crucial to enabling D2M in a mobile form factor.

MediaTek's MT6261 SoC provided a dependable baseband platform. Integrating the SL3000's broadcast capabilities with the MT6261 required deep hardware and software collaboration between Lava's R&D team and Tejas engineers. The software runs efficiently on devices with under a 250 MHz single-core processor and just 4MB RAM/Flash, ensuring smooth performance on low-resource devices—unlike many global solutions that need more power and memory.

While global competitors target high-end smartphones, this partnership focuses on bringing D2M technology to feature and entry-level smartphones, ensuring broader accessibility and bridging the digital divide. By emphasizing lightweight processing and seamless integration, Saankhya Labs, Lava, and MediaTek have secured a unique spot in the evolving mobile media landscape.

Q How exactly does D2M enable infotainment and emergency alerts without data connection?

Parag: D2M uses terrestrial broadcast airwaves (UHF band 470–582 MHz) to transmit audio, video, and text directly to mobile devices. It works like digital TV but is optimized for mobile reception via a dedicated

UHF antenna and Saankhya's SL3000 chip. This enables phones to receive live TV, OTT video, educational content, and government alerts without internet. It also supports targeted content and emergency alerts during internet outages or network failures, making it vital for public safety and disaster communication.

Q What does this mean for 400-500 million Indians still on 2G?

Parag: Nearly half a billion Indians still rely on 2G networks because mobile data costs remain unaffordable for many. For providers, offering data services at such low prices isn't viable. D2M provides a compelling solution, delivering curated video content and data services affordably for users and sustainably for providers.

Sanjeev: The inclusion of Lava's own accessibility-first features, like the BOL (text-to-speech) system and UPI integration in feature phones, complements D2M by enhancing user experience for seniors,

the visually impaired, and first-time digital users. Together, these innovations mark a pivotal moment in India's journey toward inclusive, equitable technology access.

Q What does Lava see as the long-term play for D2M?

Sanjeev: D2M's impact goes beyond internet-free content delivery. It helps reduce cellular network congestion, especially as video dominates mobile data traffic. With smartphones everywhere and content consumption rising, telecom operators face growing challenges to meet bandwidth demands cost-effectively.

By offloading high-data-use applications like live streaming, firmware



updates, and even software patches onto a broadcast channel, D2M helps unburden cellular networks. It functions like a parallel pipeline, efficient, scalable, and spectrum-friendly. Lava envisions this technology becoming integral to India's future content delivery infrastructure, especially in hybrid 5G-broadcast environments.

Q What does this innovation say about India's ambition to become a manufacturing hub?

Sanjeev: The D2M-enabled feature phone symbolizes India's growing innovation ecosystem and national ambition—not just to assemble for global brands but to design and commercialize original technologies addressing India-specific challenges. From chipset to software, design to testing, every layer is led by Indian engineering and manufacturing.

This is not a mere adaptation of existing foreign technologies but a true case of indigenous innovation with export potential. It showcases the capabilities of India's semiconductor and electronics ecosystem, exemplified by Saankhya Labs, MediaTek India, and Lava's own R&D team. It also demonstrates our readiness to lead in the global conversation on convergence technologies, especially where 5G, broadcasting, and public service delivery intersect.

Launching this product in India first and building the entire supply chain domestically redefines India's role, from low-cost manufacturing hub to a cradle of deep-tech innovation. It boldly declares that India is ready not just to make, but to invent in India. **[d]**



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